NGO CHI NGHIA

AI Engineer | GenAI, LLM & RAG

Can Tho, Vietnam · Open to Relocation and On-site Work

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Education

Nam Can Tho University, Can Tho, Vietnam. Bachelor of Engineering in Information Technology, GPA 3.44/4.00. Graduated Aug 2026.

Experience

AI Engineer Intern (Generative AI), ADHIGHTECH (Tien Phong Engineering Technology Co., Ltd.), Can Tho, Vietnam. Mar - Jul 2026.

- Implemented four retrieval approaches (dense, BM25, hybrid, cluster-based) for Vietnamese economic QA, including energy-distance cluster ranking and LLM-based relevance filtering.
- Built a FastAPI REST service for the LLM-based QA pipeline, containerized with Docker Compose on Ubuntu Linux.

Selected Projects

ERC-RAG: Energy-Based Retrieval Clustering for RAG | GitHub Mar - Jul 2026

- Designed a two-stage retrieval framework: top-40 candidates via multilingual-E5 and cosine similarity, K-Means clustering with silhouette-based selection, energy-distance cluster ranking, then LLM relevance filtering before answer generation.
- Evaluated on VNEIQAD (1,000 auto-generated QA pairs over 120,165 VEID chunks) with GPT-5.0 mini: MRR 0.62, NDCG@5 0.83, ROUGE-2 F1 0.68, BLEU-2 0.65.
- Improved BLEU-2 from 0.61 to 0.65 and ROUGE-2 F1 from 0.49 to 0.68 over the group's prior energy-based reranking RAG (different LLM, GPT-4o-mini).

Vietnamese Legal Document RAG Web Application | GitHub Jan - Mar 2026

- Built a RAG application over 6 Vietnamese legal and IT/cybersecurity documents (1,035 chunks), integrating document processing, retrieval and LLM answer generation with LangGraph. Deployed with Docker and Docker Compose.
- Ran a preliminary retrieval evaluation on 30 questions auto-generated from article titles: the expected source file and article appeared in the top-20 merged results (Chroma + BM25) for 27/30 questions (90%).

Publications & Awards

- **Co-author,** "ERC-RAG: An Energy-based Retrieval Clustering for RAG in Vietnamese Economy Chatbots." Submitted to ICEIR 2026, under review. Contributed to idea development, implementation and experiments.
- **Finalist,** FPTU Can Tho Olympic AI, May 2026.

Skills

Programming: Python
LLM: OpenAI API, Gemini API, Groq API, RAG, prompt engineering
Retrieval: multilingual-E5, ChromaDB, BM25, hybrid retrieval, Reciprocal Rank Fusion (RRF), K-Means, energy distance
Evaluation: MRR, NDCG@5, ROUGE-2, BLEU-2
Backend & tools: FastAPI, LangChain, LangGraph, Docker, Docker Compose, Ubuntu Linux, Git, GitHub
Data: NumPy, Pandas
AI-assisted development: Claude Code, OpenAI Codex CLI
Spoken Languages: Vietnamese (Native), English (VSTEP B1)